10TH CLASS BIOLOGY — NEW SYLLABUS (PECTAA)

Chapter 7 — Inheritance

Solved Exercise | New Book (PECTAA)

Exercise (Solved)

A. Select the correct answers for the following questions.

1. Chromosomes of eukaryotes are made of:

(a) Lipids and proteins
(b) Lipids and DNA
(c) RNA and proteins
(d) DNA and proteins

Answer: (d) — In eukaryotes the chromatin is made of DNA and special proteins called histones. A long molecule of DNA is wrapped around the bundles of histones, and during cell division the chromatin coils to form chromosomes.

2. What is the relationship between a gene and an allele?

(a) Alleles are alternative forms of a gene
(b) Genes are alternative forms of an allele
(c) Genes mask the effects of genes
(d) Alleles and genes are unrelated

Answer: (a) — A gene is a segment of DNA which has the information for a particular characteristic. The different forms of the same gene are called its alleles; for example, the gene of hair colour can have one allele which makes pigments and another which does not.

3. Which molecule is formed during transcription?

(a) DNA
(b) Protein
(c) mRNA
(d) Ribosome

Answer: (c) — In transcription the segment of DNA (gene) acts as a template and a type of RNA called messenger RNA (mRNA) is synthesised according to this template. Protein is formed in translation.

4. The central dogma of molecular biology is best represented by:

(a) DNA + RNA = Protein
(b) DNA → RNA + Protein
(c) DNA → RNA → Protein
(d) Protein → RNA → DNA

Answer: (c) — The information of DNA is first copied into mRNA (transcription) and the mRNA is then used to make a protein (translation). So the working of a gene, called the central dogma, is DNA → mRNA → protein.

5. What happens during transcription?

(a) A new polypeptide is created
(b) RNA is scanned by a ribosome
(c) A new copy of DNA is produced
(d) A copy of RNA is made

Answer: (d) — Transcription means the making of a copy of RNA (mRNA) according to the template of DNA. The scanning of mRNA by a ribosome happens in translation.

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6. What is the role of DNA in protein synthesis?

(a) Produces amino acids
(b) Provides the template for mRNA formation
(c) Transports proteins
(d) Stores proteins

Answer: (b) — The gene, a segment of DNA, acts as a template for the formation of mRNA. The information carried by mRNA then decides the sequence of amino acids in the protein.

7. The allele that is expressed, when two separate alleles are inherited:

(a) Dominant
(b) Recessive
(c) Homozygous
(d) Heterozygous

Answer: (a) — In a heterozygous genotype one allele may mask the working of the other. The allele which shows its effect is called dominant and the masked allele is called recessive.

8. Which of these is a reason why pea plants were suitable for Mendel's experiments?

(a) Long life cycle
(b) Easy to cross-pollinate
(c) Few seeds produced per plant
(d) No variation in traits

Answer: (b) — Pea plants have a short generation time, seven easily distinguishable characteristics and self-pollination normally occurs in their flowers, but cross-pollination can also be performed by removing the stamens and transferring the pollen. So they were easy to cross-pollinate for the experiments.

9. What does Mendel's Law of Segregation state?

(a) Genes are linked together
(b) Each allele separates during gamete formation
(c) Dominant traits appear in all generations
(d) Alleles blend together

Answer: (b) — The law of segregation states that the alleles are separated during gamete formation and each gamete receives one or the other allele, but not both. When fertilization occurs, the offspring again has two alleles.

10. In peas, allele for purple flower (P) is dominant over the allele for white flower (p). If a white flower is crossed with a heterozygous purple flower (Pp), what will be the ratio of offspring?

(a) 100 % purple
(b) 75 % purple and 25 % white
(c) 50 % purple and 50 % white
(d) 100 % white

Answer: (c) — The white flower is pp and the heterozygous purple flower is Pp. Their gametes are p from the white flower and P or p from the purple flower, which give Pp (purple) and pp (white) offspring in equal numbers. So the ratio is 1 : 1, that is 50 % purple and 50 % white.

B. Write short answers.

1. Write down the combination of alleles in the gametes of a plant, which has genotype Rryy.

The genotype of the plant is Rryy, which means it is heterozygous for the seed-shape gene (Rr) and homozygous for the seed-colour gene (yy). During gamete formation the alleles of each pair separate, and each gamete receives one allele of each gene. So the gametes of this plant can have only two combinations of alleles: **Ry** and **ry**. No gamete can contain R with r, or y with y, because one allele of each pair goes to each gamete.

2. State Mendel's law of segregation.

Mendel's law of segregation states that “the alleles are separated during gamete formation and each gamete receives one or the other allele, but not both”. Each parent has two alleles of a gene, but it passes only one of them to its offspring. When fertilization occurs, the two alleles come together again in the offspring, so the offspring again has a pair of alleles.

3. State Mendel's law of independent assortment.

Mendel's law of independent assortment states that “alleles separate independently of one another during the formation of gametes”. It means that the alleles of one gene are distributed into the gametes without any relation to the alleles of another gene. For example, in the dihybrid cross RrYy the four types of gametes RY, Ry, rY and ry are formed in equal numbers.

4. Define monohybrid cross and a dihybrid cross and give an example of each.

Monohybrid cross: it is a cross in which only one characteristic is studied. For example, Mendel crossed a true-breeding tall plant (TT) with a true-breeding short plant (tt); all the F_1 plants were tall (Tt) and the F_2 generation showed 75 % tall and 25 % short plants.

Dihybrid cross: it is a cross in which the inheritance of two characteristics is studied at the same time. For example, Mendel crossed a plant with round yellow seeds (RRYY) with a plant having wrinkled green seeds (rryy); all the F_1 plants had round yellow seeds (RrYy), and the F_2 generation showed the ratio 9 : 3 : 3 : 1.

5. Draw a cross between two pea plants. One of them has round green seeds (RRyy) while the other has wrinkled yellow seeds (rrYY).

Parents: round green seeds (RRyy) × wrinkled yellow seeds (rrYY)

Gametes: the first plant gives Ry and the second plant gives rY.

F_1 generation: all the offspring receive RrYy, that is, they are heterozygous for both characteristics and all of them have **round yellow seeds**.

Parents: RRyy (round green) × rrYY (wrinkled yellow)

Gametes: Ry × rY

F_1: RrYy — all round yellow seeds

F_2 generation: when the F_1 plants (RrYy) are self-pollinated, each plant forms four types of gametes (RY, Ry, rY and ry). Sixteen combinations are obtained, which give the phenotypic ratio **9 round yellow : 3 round green : 3 wrinkled yellow : 1 wrinkled green**.

6. Differentiate between: (a) Gene and allele

Gene	Allele
It is a segment of DNA which contains the information for a hereditary character.	It is one of the alternative forms of the same gene.
It is present at a particular position (locus) on a chromosome.	The two alleles of a gene occupy the same position on the two homologous chromosomes.
Example: the gene for eye colour, the gene for hair texture.	Example: the allele for brown eyes and the allele for blue eyes.

(b) Dominant and recessive

Dominant allele	Recessive allele
It is the allele which shows its effect (is expressed) in a heterozygous genotype.	It is the allele which is masked (not expressed) in a heterozygous genotype.
It is written with a capital letter, for example T.	It is written with a small letter, for example t.
In the genotype Tt the plant is tall because T is dominant.	The effect of t can be seen only when both alleles are recessive, as in tt (short plant).

(c) Genotype and phenotype

Genotype	Phenotype
It is the combination of the alleles present in an organism.	It is the observable outcome of the genotype, in the form of a characteristic.
It cannot be seen from outside; it is written in letters such as TT, Tt or tt.	It can be seen and measured, for example a tall plant or a short plant.
Two different genotypes may give the same phenotype, for example TT and Tt both give tall plants.	The phenotype is the result of the genotype along with the environment; for example, the final height of a plant also depends on food, water and light.

(d) F_1 and F_2 generations

F_1 generation	F_2 generation
It is the first filial generation, obtained by crossing the two parents of the P generation.	It is the second filial generation, obtained by self-pollinating the plants of the F_1 generation.
In Mendel's monohybrid cross all the F_1 plants were tall (Tt).	In the F_2 generation 75 % plants were tall and 25 % were short.
It is heterozygous for the studied characteristic (Tt).	It contains both homozygous and heterozygous plants (TT, Tt and tt).

(e) Homozygous and heterozygous

Homozygous	Heterozygous
Both alleles of a gene are the same.	The two alleles of a gene are different.
It is written as TT or tt.	It is written as Tt.
It produces only one type of gamete for that gene, for example T or t only.	It produces two types of gametes for that gene, for example T and t.
It breeds true for that characteristic.	It does not breed true, so its offspring may show the recessive characteristic.

7. In the diagram of a cell, label the components and the processes.

The diagram shows the working of a gene inside a cell. The parts to be labelled are: the nucleus, the chromosome (DNA) inside the nucleus, the mRNA formed on the DNA, and the mRNA as it moves out into the cytoplasm. The processes are **transcription**, the making of mRNA from the DNA template inside the nucleus, and **translation**, in which the ribosome reads the mRNA and joins the amino acids to form the protein. The chain of small circles at the bottom represents the amino acids joined into a protein (polypeptide).

8. Label the components in the diagram of a chromosome.

The diagram shows a chromosome with its two identical halves. The parts to be labelled are: the **chromatid** (each of the two identical halves), the **centromere** (the point at which the two chromatids are joined), the **DNA** double helix, and the **histone proteins** around which the DNA is wrapped. A chromosome consists of DNA and histones, and the two chromatids are held together by the centromere until they separate during cell division.

C. Write answers in detail.

1. Write a note on the structure and composition of chromosome.

Composition: a chromosome is made up of chromatin material. In eukaryotes the chromatin consists of DNA and special proteins called histones. Chromatin is a thread-like material in which a long molecule of DNA is wrapped around the bundles of histones. When a cell is not dividing, its chromatin is in the form of fine threads scattered in the nucleus; during cell division the chromatin coils and makes compact structures called chromosomes.

Structure: a chromosome is rod-shaped and consists of two identical halves. Each half of a chromosome is called a chromatid, and the two chromatids are attached at a point called the centromere. The centromere holds the two chromatids together until they separate during cell division. The number of chromosomes is constant for the cells of a species; the body cells are diploid (2n), so the chromosomes are present in pairs called homologous chromosomes.

Prokaryotic chromosome: the prokaryotes have only one chromosome, which is made of a circular DNA molecule. It is not bound by a nuclear envelope and is present in the cytoplasm.

2. Why did Mendel choose pea plants for his genetic experiments?

Gregor Mendel was an Austrian monk who performed experiments on garden pea from 1856 to 1863. He selected the garden pea for his experiments for several good reasons.

(i) Pea plants have a relatively short generation time, so many generations could be studied in a few years.

(ii) The pea plant has seven easily distinguishable characteristics, such as round versus wrinkled seeds, and purple versus white flowers, so the results could be seen and counted easily.

(iii) Normally self-pollination occurs in pea flowers, so true-breeding varieties could be obtained easily. At the same time cross-pollination can also be performed by removing the stamens of a flower and transferring pollen grains to the flower of another plant, so controlled crosses were possible.

Mendel first got true-breeding plants for each characteristic and then performed his monohybrid and dihybrid crosses, and the results of these experiments cleared the views of heredity.

3. Describe Mendel's experiment on two-character inheritance and state the law he proposed.

In his next experiments Mendel performed dihybrid crosses, in which the inheritance of two characteristics is studied at the same time. He studied the shape of the seeds (round or wrinkled) and the colour of the seeds (yellow or green). He first grew true-breeding varieties of pea plants; one plant had round yellow seeds (RRYY) and the other had wrinkled green seeds (rryy).

Mendel crossed these true-breeding plants and all the F_1 plants produced round yellow seeds. This proved that the allele for round seeds (R) is dominant over the allele for wrinkled seeds (r), and the allele for yellow seeds (Y) is dominant over the

allele for green seeds (y). All the F_1 plants were heterozygous for both characters (RrYy).

He then allowed self-pollination in the F_1 plants and got the F_2 generation, which had four phenotypes: 9/16 round yellow, 3/16 round green, 3/16 wrinkled yellow and 1/16 wrinkled green. The F_1 plants had produced four types of gametes (RY, Ry, rY and ry), which gave sixteen combinations in the F_2 generation. This showed that the alleles R and r separated independently of the alleles Y and y.

On the basis of this experiment Mendel proposed the **law of independent assortment**, which states that “alleles separate independently of one another during the formation of gametes”.

4. Describe Mendel's Law of Segregation? Give an example.

Mendel explained that each parent has two alleles of a gene, but a parent can pass only one allele to its offspring. During gamete formation the two alleles separate, so each gamete gets only one allele of the gene. When fertilization occurs, the offspring again receives two alleles, one from each parent. This is called the law of segregation, which states that “the alleles are separated during gamete formation and each gamete receives one or the other allele, but not both”.

Example: in the pea plant the allele for tallness (T) is dominant over the allele for shortness (t). In the P generation one parent had the alleles TT and the other had tt. Each gamete of the first parent received a single T allele and each gamete of the second parent received a single t allele. On fertilization the plants of the F_1 generation received the pair Tt, so all of them were tall.

When self-pollination was allowed in the F_1 tall plants (Tt), their gametes were of two types, T and t. These gametes combined to give three kinds of offspring in the F_2 generation: 25 % TT (tall), 50 % Tt (tall) and 25 % tt (short). So 75 % of the plants were tall and 25 % were short, which proves that the alleles segregated and were not mixed.

5. Describe how DNA and RNA take part in the synthesis of a protein.

A gene is a segment of DNA which has the information for making a particular protein. The making of a protein takes place in two steps.

1. Transcription: the segment of DNA (gene) acts as a template, and a type of RNA called messenger RNA (mRNA) is synthesised according to this template. In this way DNA transfers its information to mRNA. This process takes place in the nucleus and is called transcription.

2. Translation: the mRNA then moves out of the nucleus into the cytoplasm, where a ribosome attaches to it. The ribosome reads the information present on the mRNA and joins the amino acids in the correct order, so a protein is produced. This process is called translation.

The whole idea that DNA transfers its information to mRNA and mRNA is then used to make a protein is called the central dogma, and it is written as **DNA → mRNA → protein**. In this way DNA controls the characteristics of an organism through the proteins it makes.

D. Inquisitive questions

1. How do genes control the traits and characteristics of an organism?

Every cell of the body contains the instructions for making characteristics, and this information is present in the long molecules of DNA. A gene is a segment of this DNA which carries the information for one particular protein. The cell uses this information to make proteins in two steps: the DNA serves as a template for making mRNA (transcription), and the ribosome joins the amino acids according to the mRNA to make the protein (translation). The proteins made by a cell determine its characteristics and functions; for example, the protein pigment gives colour to the eyes and the hair, and enzymes control the chemical reactions of the body. In this way genes decide the traits of an organism, and the offspring get these genes from their parents through the gametes.

2. Why can different alleles of the same gene lead to different physical traits?

Different alleles of the same gene have slightly different base sequences of DNA, so the protein made from them is also slightly different. For example, the gene of hair colour has two alleles: one allele makes the hair pigment while the other does not make the pigment, so a person with the first allele has coloured hair while the person with the second allele has light hair. In the same way, the alleles of the ear-shape gene give a free earlobe or an attached earlobe. When both alleles are present, the dominant allele produces a working protein and shows its effect, while the recessive allele is masked. That is why the different alleles of one gene give different physical traits.

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